

PERSCHOLAS GREATER BOSTON

CYBERSECURITY PROGRAM

Module 4: Access Control Lists

Hands-On Lab Workbook — Part 2

ACL Placement | VTY Lines | Spoofing | ICMP | SNMP | IPv6 ACLs
Network Defense Course | Per Scholas Greater Boston | Cybersecurity Program
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GUIDED LAB Lab 7

ACL Placement — Standard Near Destination, Extended Near Source

1. Concept Covered

- Standard ACL placement — as close to the destination as possible
- Extended ACL placement — as close to the source as possible
- Why wrong placement blocks unintended traffic
- ACL configuration guidelines — create globally first, apply to interface second

2. Scenario

Your company has three networks — HR, IT, and Finance. Management wants to block HR from reaching the Finance server. They also want to block FTP traffic from the IT network to the Finance server while allowing all other traffic. You will configure one standard ACL and one extended ACL — placed correctly on the right interface in the right direction.

3. Network Setup

Device	Interface	IP Address	Subnet Mask
Router0	GigabitEthernet0/0	192.168.10.1	255.255.255.0
Router0	GigabitEthernet0/1	192.168.20.1	255.255.255.0
Router0	GigabitEthernet0/2	192.168.30.1	255.255.255.0
HR-PC	FastEthernet0	192.168.10.10	255.255.255.0
IT-PC	FastEthernet0	192.168.20.10	255.255.255.0
Finance Server	FastEthernet0	192.168.30.100	255.255.255.0

4. Configuration Tasks

Task A — Standard ACL: Block HR from Finance server (place close to destination):

```
# Step 1 — Create the standard ACL globally first
Router(config)#access-list 10 deny 192.168.10.0 0.0.0.255
Router(config)#access-list 10 permit any

# Step 2 — Apply outbound on G0/2 — closest to Finance server (destination)
Router(config)#interface GigabitEthernet0/2
Router(config-if)#ip access-group 10 out
Router(config-if)#end
Router#write memory
```

Task B — Extended ACL: Block FTP from IT network (place close to source):

```
# Step 1 — Create the extended ACL globally first
Router(config)#ip access-list extended BLOCK-IT-FTP
Router(config-ext-nacl)#deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.100 eq 21
Router(config-ext-nacl)#deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.100 eq 20
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit

# Step 2 — Apply inbound on G0/1 — closest to IT network (source)
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip access-group BLOCK-IT-FTP in
Router(config-if)#end
Router#write memory
```

Task C — Demonstrate why wrong placement matters:

- Move access-list 10 to G0/0 inbound instead — test what happens to HR traffic going to IT network
- Observe that HR is now blocked from reaching IT as well — not just Finance
- Move it back to G0/2 outbound — confirm HR can reach IT again but not Finance

5. Verification

```
Router#show access-lists
Router#show ip interface GigabitEthernet0/0
Router#show ip interface GigabitEthernet0/1
Router#show ip interface GigabitEthernet0/2
```

6. Expected Outcome

Test	Expected Result	Pass / Fail
HR-PC pings Finance Server	Fails — standard ACL blocks HR outbound on G0/2	[] Pass [] Fail
HR-PC pings IT-PC	Succeeds — standard ACL on G0/2 does not affect this path	[] Pass [] Fail

IT-PC accesses Finance Server on port 21	Fails — extended ACL blocks FTP inbound on G0/1	☐ Pass ☐ Fail
IT-PC pings Finance Server	Succeeds — only FTP is blocked, ICMP is permitted	☐ Pass ☐ Fail
Task C — HR-PC pings IT-PC with ACL on G0/0	Fails — wrong placement blocks more than intended	☐ Pass ☐ Fail

Reflection Questions:

- Why did moving the standard ACL to G0/0 inbound block HR from reaching IT as well?
- Why is the extended ACL placed inbound on G0/1 instead of outbound on G0/2?

GUIDED LAB Lab 8

Apply ACL to Interface and VTY Lines

1. Concept Covered

- Applying ACLs to physical interfaces using ip access-group
- Applying ACLs to VTY lines using ip access-class to restrict remote SSH access
- Difference between removing an ACL from an interface and deleting it from the router
- Verifying ACL application using show ip interface

2. Scenario

Your company router needs two security controls. First — only the Management PC should be allowed to SSH into the router remotely. All other devices must be blocked from remote access. Second — a named extended ACL must be applied on the internet-facing interface to control what traffic enters the network from outside.

3. Network Setup

Device	Interface	IP Address	Subnet Mask
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0
Router0	GigabitEthernet0/1	10.0.0.1	255.255.255.0
Mgmt-PC	FastEthernet0	192.168.1.10	255.255.255.0
PC1	FastEthernet0	192.168.1.20	255.255.255.0
Internet Host	FastEthernet0	10.0.0.100	255.255.255.0

4. Configuration Tasks

Step 1 — Enable SSH on the router:

```
Router(config)#hostname R1
R1(config)#ip domain-name perscholas.org
R1(config)#crypto key generate rsa modulus 1024
R1(config)#username admin secret cisco123
R1(config)#line vty 0 4
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#exit
```

Step 2 — Create and apply a standard ACL to VTY lines using ip access-class:

```
# Create the ACL globally first
R1(config)#ip access-list standard SSH-ONLY
R1(config-std-nacl)#permit host 192.168.1.10
R1(config-std-nacl)#deny any
R1(config-std-nacl)#exit

# Apply to VTY lines using ip access-class — not ip access-group
R1(config)#line vty 0 4
R1(config-line)#access-class SSH-ONLY in
R1(config-line)#exit
```

Step 3 — Create and apply a named extended ACL on the internet-facing interface:

```
R1(config)#ip access-list extended INTERNET-IN
R1(config-ext-nacl)#permit tcp any 192.168.1.0 0.0.0.255 eq 80
R1(config-ext-nacl)#permit tcp any 192.168.1.0 0.0.0.255 eq 443
R1(config-ext-nacl)#deny ip any any
R1(config-ext-nacl)#exit

R1(config)#interface GigabitEthernet0/1
R1(config-if)#ip access-group INTERNET-IN in
R1(config-if)#end
R1#write memory
```

Step 4 — Practice removing ACL from interface without deleting it:

```
# Remove from interface only — ACL still exists on router
R1(config)#interface GigabitEthernet0/1
R1(config-if)#no ip access-group INTERNET-IN in
R1(config-if)#exit

# Verify ACL still exists
R1#show access-lists

# Delete ACL completely from router
R1(config)#no ip access-list extended INTERNET-IN
R1#show access-lists
```

5. Verification

```
R1#show access-lists
R1#show ip interface GigabitEthernet0/1
R1#show running-config | section line vty
R1#show running-config | section ip access-list
```

6. Expected Outcome

Test	Expected Result	Pass / Fail
Mgmt-PC SSH to Router	Succeeds — explicitly permitted by SSH-ONLY ACL	☐ Pass ☐ Fail
PC1 SSH to Router	Fails — blocked by deny any in SSH-ONLY ACL	☐ Pass ☐ Fail
Internet Host sends HTTP to inside network	Succeeds — port 80 permitted by INTERNET-IN	☐ Pass ☐ Fail
Internet Host sends Telnet to inside network	Fails — not permitted by INTERNET-IN	☐ Pass ☐ Fail
After no ip access-group — show access-lists	ACL still exists — only unapplied from interface	☐ Pass ☐ Fail
After no ip access-list — show access-lists	ACL completely gone from router	☐ Pass ☐ Fail

Reflection Question:

- What is the difference between ip access-group and ip access-class — and when do you use each one?

GUIDED LAB Lab 9

Mitigate IP Spoofing Attacks

1. Concept Covered

- IP address spoofing — attacker fakes source IP address to bypass security controls
- Blocking known spoofed address ranges on the internet-facing interface
- Blocking private addresses arriving from the internet — they should never appear there
- Blocking internal addresses arriving from outside — clear sign of spoofing

2. Scenario

Your router connects the internal network to the internet. You need to configure two ACLs — one on the internet-facing interface to block all known spoofed and invalid source addresses, and one on the internal interface to block any packet that claims to come from the internal network but arrives from outside.

3. Network Setup

Device	Interface	IP Address	Subnet Mask
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0
Router0	Serial0/0/0	209.165.200.1	255.255.255.252
Internal PC	FastEthernet0	192.168.1.10	255.255.255.0
Simulated Internet Host	FastEthernet0	209.165.200.2	255.255.255.252

4. Configuration Tasks

Step 1 — Create ACL to block spoofed addresses on internet-facing interface:

```
Router(config)#ip access-list extended ANTI-SPOOF-INTERNET

# Block all-zeros addresses
Router(config-ext-nacl)#deny ip 0.0.0.0 0.255.255.255 any

# Block loopback addresses — should never arrive from internet
Router(config-ext-nacl)#deny ip 127.0.0.0 0.255.255.255 any

# Block APIPA addresses — self-assigned, never cross internet
Router(config-ext-nacl)#deny ip 169.254.0.0 0.0.255.255 any

# Block RFC 1918 private ranges — never arrive from internet
Router(config-ext-nacl)#deny ip 10.0.0.0 0.255.255.255 any
Router(config-ext-nacl)#deny ip 172.16.0.0 0.15.255.255 any
Router(config-ext-nacl)#deny ip 192.168.0.0 0.0.255.255 any

# Block multicast range arriving as unicast
Router(config-ext-nacl)#deny ip 224.0.0.0 15.255.255.255 any

# Permit all other legitimate traffic
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit

# Apply inbound on internet-facing interface
Router(config)#interface Serial0/0/0
Router(config-if)#ip access-group ANTI-SPOOF-INTERNET in
Router(config-if)#end
```

Step 2 — Create ACL to block internal address arriving from outside:

```
Router(config)#ip access-list extended ANTI-SPOOF-INTERNAL

# Only permit packets genuinely from the internal network
Router(config-ext-nacl)#permit ip 192.168.1.0 0.0.0.255 any

# Block anything else claiming to be from internal network
Router(config-ext-nacl)#deny ip any any
Router(config-ext-nacl)#exit

# Apply inbound on internal interface
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip access-group ANTI-SPOOF-INTERNAL in
Router(config-if)#end
Router#write memory
```

5. Verification

```
Router#show access-lists
Router#show ip interface Serial0/0/0
Router#show ip interface GigabitEthernet0/0
```

Test by sending ping from the simulated internet host using a spoofed private source address in Packet Tracer simulation mode and observe that it is blocked.

6. Expected Outcome

Test	Expected Result	Pass / Fail
Legitimate internet traffic reaches internal network	Succeeds — permit ip any any allows it through	☐ Pass ☐ Fail
Packet with source 192.168.x.x arrives from internet	Fails — RFC 1918 address blocked by ANTI-SPOOF-INTERNET	☐ Pass ☐ Fail
Packet with source 127.x.x.x arrives from internet	Fails — loopback address blocked immediately	☐ Pass ☐ Fail
Packet with source 10.x.x.x arrives from internet	Fails — RFC 1918 private range blocked	☐ Pass ☐ Fail
Internal PC sends traffic outbound	Succeeds — 192.168.1.0/24 permitted by ANTI-SPOOF-INTERNAL	☐ Pass ☐ Fail

Reflection Question:

- Why would a packet claiming to come from 192.168.1.50 but arriving on the internet-facing interface be considered spoofed?

PRACTICE LAB Lab 10

Permit Necessary Services Through a Firewall

1. Concept Covered

- Explicit permit strategy — deny everything by default, permit only what is needed
- Permitting DNS, SMTP, FTP, SSH, Syslog, and SNMP through a firewall ACL
- Restricting administrative access to only specific management hosts
- Applying the ACL on the correct interface and direction

2. Scenario

You are setting up ACL rules on the router's internet-facing interface. Management has given you a list of services that must be allowed through the firewall and a list of services that must be blocked. Your task is to build the ACL using the explicit permit strategy — only permit what is listed and deny everything else.

3. Network Setup

Device	Interface	IP Address	Subnet Mask
Router0	Serial0/0/0	209.165.200.1	255.255.255.252
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0
DNS Server	FastEthernet0	192.168.1.50	255.255.255.0
Mail Server	FastEthernet0	192.168.1.60	255.255.255.0
Mgmt Station	FastEthernet0	10.0.0.5	255.255.255.0

4. Configuration Tasks

Build an extended ACL called FIREWALL-POLICY applied inbound on Serial0/0/0 that:

- 1. Permits DNS (UDP port 53) traffic to the DNS server at 192.168.1.50
- 2. Permits SMTP (TCP port 25) traffic to the mail server at 192.168.1.60
- 3. Permits FTP (TCP ports 21 and 20) traffic to the mail server
- 4. Permits SSH (TCP port 22) only from the management station at 10.0.0.5 to the router
- 5. Permits Syslog (UDP port 514) from the router interface only
- 6. Permits SNMP traps (UDP port 162) from the management station only
- 7. Denies all other traffic implicitly

Hint: Use the following command structure as a starting point:

```
Router(config)#ip access-list extended FIREWALL-POLICY
Router(config-ext-nacl)#permit _____ any host _____ eq _____
Router(config-ext-nacl)#permit _____ any host _____ eq _____
# Continue for all required services
Router(config-ext-nacl)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip access-group FIREWALL-POLICY in
```

5. Verification

- Use show access-lists to confirm all ACEs are configured correctly
- Use show ip interface Serial0/0/0 to confirm the ACL is applied inbound
- Test each service in Packet Tracer simulation mode and record which ones pass and which ones fail

6. Expected Outcome

Test	Expected Result	Pass / Fail
DNS traffic to 192.168.1.50 port 53	Succeeds — DNS explicitly permitted	☐ Pass ☐ Fail
SMTP traffic to 192.168.1.60 port 25	Succeeds — SMTP explicitly permitted	☐ Pass ☐ Fail
SSH from 10.0.0.5 to router	Succeeds — management station permitted for SSH	☐ Pass ☐ Fail

SSH from any other host to router	Fails — only management station is permitted	☐ Pass ☐ Fail
HTTP traffic from internet to inside	Fails — HTTP not in the permitted services list	☐ Pass ☐ Fail
Telnet from internet to any inside host	Fails — Telnet not in the permitted services list	☐ Pass ☐ Fail

Reflection Question:

- Why is it safer to explicitly permit only the services you need rather than explicitly deny only the services you want to block?

PRACTICE LAB Lab 11

Mitigate ICMP and SNMP Attacks

1. Concept Covered

- Controlling specific ICMP message types — permitting only what is needed
- Blocking inbound ICMP echo while allowing inside users to ping and receive replies
- Disabling SNMP on devices that do not require it using no snmp-server
- Restricting SNMP access to authorized management hosts only

2. Scenario

Your organization wants to control ICMP traffic tightly. Inside users should be able to ping external hosts and receive replies — but external hosts should not be able to ping inside devices directly. You also need to secure SNMP — disable it on devices that do not need it, and restrict it to only the management station on devices that do require it.

3. Network Setup

Device	Interface	IP Address	Subnet Mask
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0
Router0	Serial0/0/0	209.165.200.1	255.255.255.252
Inside PC	FastEthernet0	192.168.1.10	255.255.255.0
Mgmt Station	FastEthernet0	192.168.1.50	255.255.255.0
Internet Host	FastEthernet0	209.165.200.2	255.255.255.252

4. Configuration Tasks

Task A — Configure ICMP ACLs on both interfaces:

- On Serial0/0/0 inbound — permit echo-reply, source-quench, and unreachable — deny all other ICMP — permit all non-ICMP traffic
- On GigabitEthernet0/0 inbound — permit echo, parameter-problem, packet-too-big, and source-quench — deny all other ICMP — permit all non-ICMP traffic

Task B — Disable SNMP on the router:

```
# Completely disable SNMP if not required
Router(config)#no snmp-server
```

Task C — If SNMP is required — restrict it to the management station only:

```
# Create ACL to permit SNMP only from management station
Router(config)#ip access-list standard SNMP-ACCESS
Router(config-std-nacl)#permit host 192.168.1.50
Router(config-std-nacl)#deny any
Router(config-std-nacl)#exit

# Apply ACL to SNMP community string
Router(config)#snmp-server community public ro SNMP-ACCESS
```

5. Verification

- Use show access-lists to confirm ICMP ACEs are configured
- Test ping from Inside PC to Internet Host — should succeed
- Test ping from Internet Host to Inside PC — should fail
- Use show snmp to verify SNMP status after disabling it

6. Expected Outcome

Test	Expected Result	Pass / Fail
Inside PC pings Internet Host	Succeeds — echo permitted outbound from inside	☐ Pass ☐ Fail
Internet Host pings Inside PC	Fails — echo blocked inbound on Serial0/0/0	☐ Pass ☐ Fail
Inside PC receives ping reply from internet	Succeeds — echo-reply permitted inbound	☐ Pass ☐ Fail
SNMP query from Mgmt Station to router	Succeeds — management station is permitted	☐ Pass ☐ Fail
SNMP query from any other host	Fails — blocked by SNMP-ACCESS ACL	☐ Pass ☐ Fail
show snmp after no snmp-server	SNMP is completely disabled — no response to any query	☐ Pass ☐ Fail

Reflection Question:

- Why is it not safe to completely block all ICMP traffic — what would break on the network if you did?

IPv6 ACL Configuration — Challenge Lab

1. Concept Covered

- IPv6 ACL creation using ipv6 access-list — all IPv6 ACLs must be named
- Applying IPv6 ACLs using ipv6 traffic-filter — not ip access-group
- Implicit deny ipv6 any any at the bottom of every IPv6 ACL
- Explicitly permitting NDP (Neighbor Discovery Protocol) traffic — nd-na and nd-ns
- Filtering IPv6 traffic by source prefix, destination, and protocol

2. Scenario

Your organization is running a dual-stack network — both IPv4 and IPv6 are active. You must configure IPv6 ACLs to secure IPv6 traffic on the network. The security policy requires you to permit only the internal IPv6 network to send traffic outbound, block all other IPv6 traffic, and ensure NDP continues to work correctly so devices can discover each other.

3. Network Setup

Device	Interface	IP Address	Subnet Mask
Router0	GigabitEthernet0/0	2001:DB8:ACAD:1::1/64	N/A — IPv6 uses prefix length
Router0	GigabitEthernet0/1	2001:DB8:ACAD:2::1/64	N/A — IPv6 uses prefix length
Inside PC	FastEthernet0	2001:DB8:ACAD:1::10/64	N/A
Outside Host	FastEthernet0	2001:DB8:ACAD:2::100/64	N/A

4. Your Tasks

For each requirement below — determine and configure the correct IPv6 ACL:

1. Create a named IPv6 ACL called LAN-ONLY that permits traffic from 2001:DB8:ACAD:1::/64 to any destination
2. Explicitly permit NDP neighbor advertisement messages — nd-na
3. Explicitly permit NDP neighbor solicitation messages — nd-ns
4. Deny all other IPv6 traffic
5. Apply the ACL inbound on GigabitEthernet0/0 using ipv6 traffic-filter
6. Verify using show ipv6 access-lists and show ipv6 interface GigabitEthernet0/0

Hints (use only if needed):

- The command to create an IPv6 ACL is ipv6 access-list — not ip access-list
- The command to apply is ipv6 traffic-filter — not ip access-group
- NDP uses ICMPv6 — use permit icmp any any nd-na and permit icmp any any nd-ns
- If you forget to permit NDP — devices will not be able to discover each other

5. Verification

```
Router#show ipv6 access-lists
Router#show ipv6 interface GigabitEthernet0/0
Router#show running-config | section ipv6 access-list
```

6. Expected Outcome

Test	Expected Result	Pass / Fail
Inside PC sends IPv6 traffic outbound	Succeeds — 2001:DB8:ACAD:1::/64 is permitted	☐ Pass ☐ Fail
Outside Host sends IPv6 traffic inbound	Fails — not in the permitted prefix — implicit deny	☐ Pass ☐ Fail
NDP works between devices on same segment	Succeeds — nd-na and nd-ns explicitly permitted	☐ Pass ☐ Fail
show ipv6 access-lists shows LAN-ONLY	ACL visible with all four ACEs and sequence numbers	☐ Pass ☐ Fail
show ipv6 interface G0/0	LAN-ONLY listed as inbound IPv6 access list	☐ Pass ☐ Fail

Documentation Space — Write your IPv6 ACL configuration here:

Write your IPv6 ACL commands here:

Reflection Questions:

- What happens to IPv6 communication on the network if you configure deny ipv6 any any without first permitting NDP traffic?
- Why does IPv6 use ipv6 traffic-filter instead of ip access-group to apply ACLs to an interface?
- What is the key difference between ip access-list and ipv6 access-list in terms of naming requirements?

End of Module 4 Lab Workbook — Part 2

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